

Foundation (Jalapeno)

Q1. Factor the expression: $x^2 + 3x + 2x + 6$

- (A) $(x + 3)(x + 2)$
- (B) $(x + 2)(x + 3)$
- (C) $(x + 1)(x + 6)$
- (D) $(x + 6)(x + 1)$
- (E) $(x + 2)(x + 1)$

Q2. Factor the expression: $x^2 + 5x + x + 5$

- (A) $(x + 5)(x + 1)$
- (B) $(x + 1)(x + 5)$
- (C) $(x + 6)(x - 1)$
- (D) $(x - 1)(x + 6)$
- (E) $(x + 2)(x + 3)$

Q3. Factor the expression: $2x^2 + 7x + 3x + 10$

- (A) $(2x + 3)(x + 5)$
- (B) $(2x + 5)(x + 3)$
- (C) $(2x + 2)(x + 5)$
- (D) $(x + 5)(2x + 2)$
- (E) $(2x + 10)(x - 1)$

Q4. Factor the expression: $3x^2 + 2x + x + 10$

- (A) $(3x + 2)(x + 5)$
- (B) $(3x + 1)(x + 10)$
- (C) $(3x + 10)(x + 1)$
- (D) $(x + 10)(3x + 1)$
- (E) $(3x + 5)(x + 2)$

Q5. Factor the expression: $x^2 + 2x + 5x + 10$

- (A) $(x + 5)(x + 2)$
- (B) $(x + 2)(x + 5)$
- (C) $(x + 10)(x - 1)$
- (D) $(x + 1)(x + 10)$
- (E) $(x - 1)(x + 10)$

Q6. Factor the expression: $2x^2 + 9x + 4x + 18$

- (A) $(2x + 9)(x + 2)$
- (B) $(2x + 4)(x + 9)$
- (C) $(2x + 4)(x + 9)$
- (D) $(2x + 6)(x + 3)$
- (E) $(2x + 8)(x + 9/2)$

Q7. Factor the expression: $x^2 + x + 4x + 4$

- (A) $(x + 4)(x + 1)$
- (B) $(x + 2)(x + 2)$
- (C) $(x + 3)(x + 4)$
- (D) $(x + 4)(x - 1)$
- (E) $(x^2 + 1)$

Q8. Factor the expression: $x^2 + 6x + 2x + 12$

- (A) $(x + 6)(x + 2)$
- (B) $(x + 2)(x + 6)$
- (C) $(x + 4)(x + 3)$
- (D) $(x + 3)(x + 4)$
- (E) $(x + 1)(x + 12)$

Open Ended Questions (FRQ)

Q9. Factor the expression: $3x^2 + 10x + 2x + 8$

Q10. Verify the factoring of $x^2 + 5x + 6$ by multiplication.

Chef's Tips

- Tip 1: Check each student's work carefully for accuracy.
- Tip 2: Ensure that students have the necessary materials and space to complete the test.
- Tip 3: Make sure to keep the test secure and confidential.